

From Specialists to Generalists: Merging LoRA-Adapted Models with DARE and AdaMerging

Mathis Richard | 346419 | mathis.richard@epfl.ch

Max Henrotin | 341463 | max.henrotin@epfl.ch

Morgane Magnin | 347041 | morgane.magnin@epfl.ch

Julien Erbland | 346893 | julien.erbland@epfl.ch

Émainèlpé

1 Literature Review

This section surveys the literature on post-training across four distinct domains: mathematical reasoning, general knowledge, multilinguality, and safety.

1.1 Mathematical Reasoning

For small models, diversity matters more than dataset size. [Yuan et al. \(2023\)](#) filter for correct solutions via Rejection Sampling Fine-Tuning, and [Yu et al. \(2024b\)](#) expand question variety by rewriting and reversing existing problems. The dataset from [Toshniwal et al. \(2024\)](#) combine both ideas. [Tong et al. \(2024\)](#) show that allocating more effort to harder problems improves results, while [Dang and Ngo \(2025\)](#) find that mixing easy and hard problems stabilizes early RL training on small models. Building on this, we target the band where it has the most to learn between 20% and 80% of difficulty for the model.

1.2 General Knowledge

While standard SFT for injecting new facts often causes hallucinations ([Gekhman et al., 2024](#)), prompting models to explicitly "think" improves factual extraction on general knowledge MCQs ([Mitra et al., 2023](#)). To enforce this reasoning, we adopt Reinforcement Learning with Verifiable Rewards (RLVR) ([Lambert et al., 2024](#)), leveraging exact-match as a cost-free, rule-based reward signal. We apply SFT and RLVR via LoRA to the separated train sets of MMLU ([Hendrycks et al., 2021](#)) and ARC-Challenge ([Clark et al., 2018](#)), enhancing factual capabilities without test contamination.

1.3 Multilinguality

Cross-Lingual Reasoning in low-resource languages often causes language models to misunderstand problems and fail logically. [Khanh-Tung Tran \(2025\)](#) propose English-Pivoted CoT Training, which trains models to generate their internal CoT in English, while outputting the final

response in the target language. High-quality data is scarce outside of English. [Josh Barua \(2026\)](#), show that fine-tuning on reasoning traces translated from gold English data outperforms fine-tuning on target-language distilled directly from large models. For smaller models even a few samples is enough for good performance. To validate the model's cross-lingual reasoning consistency, we can use benchmarks such as MGSM and MSVAMP, [Lai and Nissim \(2024\)](#), while mCoT-MATH can be used for broader instructions.

1.4 Safety

We use Direct Preference Optimization (DPO) ([Rafailov et al., 2023](#)) for safety alignment, as it avoids the cost of full RLHF (reward modeling + PPO) by directly optimizing chosen over rejected responses, making it well-suited for small-model post-training. SafeDPO ([Kim et al., 2025](#)) further adapts this objective for safety, and prior work shows preference optimization significantly improves LLM safety ([Alami et al., 2024](#)). PKU-SafeRLHF ([Ji et al., 2024](#)) provides preference pairs suitable for DPO-based training. Evaluation can be performed on the English subset of Safety-Bench ([Zhang et al., 2023](#)), with XSTest ([Röttger et al., 2023](#)) used as a diagnostic suite to detect over-refusal on benign prompts.

1.5 Model Merging

To combine the four LoRA adapters, we apply DARE ([Yu et al., 2024a](#)) followed by AdaMerging ([Yang et al., 2024](#)), with TIES ([Yadav et al., 2023](#)) as a fallback. DARE prunes low-magnitude weights, while AdaMerging learns layer-wise combination weights via entropy minimization on unlabeled data. If interference persists, TIES resolves conflicts by trimming low-magnitude updates, selecting a dominant sign per parameter via majority vote, and averaging only sign-consistent contributions.

References

- Reda Alami, Ali Khalifa Almansoori, Ahmed Alzubaidi, Mohamed El Amine Seddik, Mugariya Farooq, and Hakim Hacid. 2024. [Alignment with preference optimization is all you need for LLM safety](#). *arXiv preprint arXiv:2409.07772*.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. [Think you have solved question answering? try arc, the ai2 reasoning challenge](#). *arXiv preprint arXiv:1803.05457*.
- Quy-Anh Dang and Chris Ngo. 2025. Reinforcement learning for reasoning in small LLMs: What works and what doesn't. *arXiv preprint arXiv:2503.16219*.
- Zorik Gekhman, Gal Yona, Roei Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Heri Ramampiaro. 2024. [Does fine-tuning llms on new knowledge encourage hallucinations?](#)
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). In *International Conference on Learning Representations (ICLR)*.
- Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Juntao Dai, Boren Zheng, Tianyi Qiu, Jiayi Zhou, Kaile Wang, Boxuan Li, Sirui Han, Yike Guo, and Yaodong Yang. 2024. [PKU-SafeRLHF: Towards multi-level safety alignment for LLMs with human preference](#). *arXiv preprint arXiv:2406.15513*.
- Kayo Yin Alane Suhr Josh Barua, Seun Eisape. 2026. Long chain-of-thought reasoning across languages. <https://arxiv.org/abs/2508.14828>.
- Hoang D. Nguyen Khanh-Tung Tran, Barry O'Sullivan. 2025. Scaling test-time compute for low-resource languages: Multilingual reasoning in llms. <https://arxiv.org/html/2504.02890v1>.
- Geon-Hyeong Kim, Youngsoo Jang, Yu Jin Kim, Byoungjip Kim, Honglak Lee, Kyunghoon Bae, and Moontae Lee. 2025. [SafeDPO: A simple approach to direct preference optimization with enhanced safety](#). *arXiv preprint arXiv:2505.20065*.
- Huiyuan Lai and Malvina Nissim. 2024. mcot: Multilingual instruction tuning for reasoning consistency in language models. <https://arxiv.org/abs/2406.02301>.
- Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, et al. 2024. [Tulu 3: Pushing frontiers in open language model post-training](#). *arXiv preprint arXiv:2411.15124*.
- Arindam Mitra, Luciano Del Corro, Shweti Mahajan, Andres Codash, Peter Simeth, Corby Rosset, Guoqing Zheng, Ahmed Awadallah, et al. 2023. [Orca 2: Teaching small language models how to reason](#).
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. [Direct preference optimization: Your language model is secretly a reward model](#). *arXiv preprint arXiv:2305.18290*.
- Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2023. [XSTest: A test suite for identifying exaggerated safety behaviours in large language models](#). *arXiv preprint arXiv:2308.01263*.
- Yuxuan Tong, Xiwen Zhang, Rui Wang, Ruidong Wu, and Junxian He. 2024. DART-Math: Difficulty-aware rejection tuning for mathematical problem-solving. *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2407.13690.
- Shubham Toshniwal, Wei Du, Ivan Moshkov, Branislav Kisacanin, Alexan Ayrapetyan, and Igor Gitman. 2024. OpenMathInstruct-2: Accelerating AI for math with massive open-source instruction data. *arXiv preprint arXiv:2410.01560*.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. 2023. [TIES-Merging: Resolving interference when merging models](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. 2024. AdaMerging: Adaptive model merging for multi-task learning. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.02575.
- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. 2024a. Language models are super mario: Absorbing abilities from homologous models as a free lunch. *Proceedings of the 41st International Conference on Machine Learning (ICML)*. ArXiv:2311.03099 (DARE).
- Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T. Kwok, Zhengguo Li, Adrian Weller, and Weiyang Liu. 2024b. MetaMath: Bootstrap your own mathematical questions for large language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2309.12284.
- Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. 2023. Scaling relationship on learning mathematical reasoning with large language models. *arXiv preprint arXiv:2308.01825*.
- Zhexin Zhang, Leqi Lei, Lindong Wu, Rui Sun, Yongkang Huang, Chong Long, Xiao Liu, Xuanyu Lei, Jie Tang, and Minlie Huang. 2023. [Safety-Bench: Evaluating the safety of large language models with multiple choice questions](#). *arXiv preprint arXiv:2309.07045*.